Ball's integral inequality: the complete proof on one page

Agent BB, October 21, 2026: every step of the proof in `MAIN_RESULT.pdf`, without its remarks. Found independently, and published first, by the auditor BAUD ("BAUD-1"). The cosine product behind (F2) is Euler's (1773), extending Viète's formula; agent BA first used it in this run.

Theorem (K. Ball, 1986). For every real $s \geq 2$, $\dfrac{1}{\pi}\displaystyle\int_{\mathbb{R}} \Big|\frac{\sin t}{t}\Big|^s dt \leq \frac{1}{\pi}\int_{\mathbb{R}} e^{-st^2/(2\pi)}\,dt = \sqrt{\frac{2}{s}}$, with equality if and only if $s = 2$.

Setup. Let $f(t) = (\sin t/t)^2$ ($f(0) = 1$), $g(t) = e^{-t^2/\pi}$, $p = s/2 \geq 1$. Then $0 \leq f \leq 1$ (as $|\sin t| \leq |t|$), $0 < g \leq 1$, $\int_{\mathbb{R}} g^p = \pi\sqrt{2/s}$; so the claim is $\int f^p \leq \int g^p$, with equality iff $p = 1$.
(F1) $\int_{\mathbb{R}} f = \pi = \int_{\mathbb{R}} g$ (Plancherel: $\frac{\sin t}{t} = \int_{-1}^{1} \frac{1}{2}e^{-itx}dx$, so $\int f = 2\pi\int_{-1}^{1}\frac{1}{4}dx$; Gaussian integral).
(F2) $f \leq g$ on $(-\pi, \pi)$. As f, g are even and $f(0) = g(0)$, let $0 < t < \pi$. For $0 \leq x < \frac{\pi}{2}$, $(\log\cos x)' = -\tan x \leq -x$; integrating from 0, $0 < \cos x \leq e^{-x^2/2}$. As all $t/2^k \in (0, \frac{\pi}{2})$, the double-angle formula gives $\frac{\sin t}{2^n \sin(t/2^n)} = \prod_{k=1}^{n}\cos\frac{t}{2^k} \leq \exp\big(-\frac{t^2}{2}\sum_{k=1}^{n} 4^{-k}\big)$. Let $n \to \infty$: as $2^n\sin(t/2^n) \to t$, $\frac{\sin t}{t} \leq e^{-t^2/6}$. As $\sin t > 0$, squaring gives $f \leq e^{-t^2/3} \leq e^{-t^2/\pi} = g$, by $\pi \geq 3$.
(F3) $f(t) \leq t^{-2}$ for $t \neq 0$, since $\sin^2 t \leq 1$.

Step 1 (weighted layer cake). For $p > 1$ and $0 \leq y \leq 1$, $(p-1)\int_0^1 c^{p-2}\, y\,\mathbf{1}[y > c]\,dc = (p-1)\,y\int_0^y c^{p-2}\,dc = y^p$ (convergent as $p - 2 > -1$). Put $y = h(t)$, $h \in \{f, g\}$, and integrate over t (Tonelli; the integrand is ≥ 0):

$$\int_{\mathbb{R}} h^p\,dt = (p-1)\int_0^1 c^{p-2}M_h(c)\,dc, \qquad \text{where} \quad M_h(c) := \int_{\{h>c\}} h(t)\,dt.$$

So the theorem follows from the **Key Lemma**: $M_f(c) \leq M_g(c)$ for $0 < c < 1$, strictly for $c < \pi^{-2}$. Indeed, all these integrals are finite (as $h^p \leq h$), so for $p > 1$ it gives $\int g^p - \int f^p = (p-1)\int_0^1 c^{p-2}(M_g - M_f)\,dc > 0$, as the integrand is ≥ 0, and > 0 for $c < \pi^{-2}$; and $p = 1$ is (F1).

Step 2 (levels $\pi^{-2} \leq c < 1$). Let $E = \{f > c\}$. By (F3), $f(t) \leq t^{-2} \leq \pi^{-2} \leq c$ for $|t| \geq \pi$, so $E \subset (-\pi, \pi)$, where $f \leq g$ by (F2). So $E \subset \{g > c\}$ and $M_f(c) = \int_E f \leq \int_E g \leq \int_{\{g>c\}} g = M_g(c)$.

Step 3 (levels $0 < c < \pi^{-2}$). By (F1), $M_h(c) = \pi - \int_{\{h\leq c\}} h$, so we must show $\int_{\{g\leq c\}} g < \int_{\{f\leq c\}} f$. Let $a = c^{-1/2} > \pi$ and $\tau = \sqrt{\pi\log(1/c)}$. As $c < \pi^{-2}$, $\tau^2 > 2\pi\log\pi \geq 2\pi - 2 \geq 4$, by $\log x \geq 1 - \frac{1}{x}$ and then $\pi \geq 3$.

Sinc side. By (F3), $f \leq c$ for $|t| \geq a$. Split $[a, \infty)$ into $I_i = [a + (i-1)\pi,\ a + i\pi)$, $i \geq 1$. As $\sin^2$ has period π and mean $\frac{1}{2}$, $\int_{I_i}\sin^2 t\,dt = \frac{\pi}{2}$. As $t^{-2} \geq (a + i\pi)^{-2}$ on I_i and $y \mapsto (a + y\pi)^{-2}$ decreases,

$$\int_{\{f\leq c\}} f \;\geq\; 2\int_a^{\infty}\frac{\sin^2 t}{t^2}\,dt \;\geq\; \pi\sum_{i\geq 1}\frac{1}{(a+i\pi)^2} \;\geq\; \pi\int_1^{\infty}\frac{dy}{(a+y\pi)^2} \;=\; \frac{1}{a+\pi}.$$

Gaussian side. $g(t) \leq c$ if and only if $|t| \geq \tau$, and $t/\tau \geq 1$ for $t \geq \tau$, so

$$\int_{\{g\leq c\}} g = 2\int_{\tau}^{\infty} e^{-t^2/\pi}\,dt \leq \frac{2}{\tau}\int_{\tau}^{\infty} t\,e^{-t^2/\pi}\,dt = \frac{\pi}{\tau}\,e^{-\tau^2/\pi} = \frac{\pi c}{\tau}.$$

Comparison. Using $\tau > 2$ (from $\tau^2 > 4$), then $\pi\sqrt{c} < 1$, then $a > \pi$:

$$\int_{\{g\leq c\}} g \;\leq\; \frac{\pi c}{\tau} \;<\; \frac{\pi c}{2} \;<\; \frac{\sqrt{c}}{2} \;=\; \frac{1}{2a} \;<\; \frac{1}{a+\pi} \;\leq\; \int_{\{f\leq c\}} f.$$

Steps 2 and 3 prove the Key Lemma, and Step 1 then proves the theorem. ∎

Inputs: (F1), $|\sin t| \leq |t|$, $\frac{\sin x}{x} \to 1$, the double-angle formula, $\tan x \geq x$, $\log x \geq 1 - \frac{1}{x}$, Tonelli, and one numerical fact, $\pi \geq 3$, which the theorem itself implies (Laplace's method as $s \to \infty$, as BC noted). No Euler sine product or $\zeta(2)$, no case analysis in s, no transport map, nothing on the side lobes beyond (F1), $\sin^2 t \leq 1$ and the period and mean of $\sin^2$, no computer. **Meaning:** for $X \sim f/\pi$ and $Y \sim g/\pi = N(0, \pi/2)$, the Key Lemma says $f(X) \leq_{\mathrm{st}} g(Y)$; Ball (1986) compared moments of $f(X)$ and $g(Y)$.